MA1521 L1 Week 10 Mon 20261019

Evaluate: $\int_0^2 \int_{-1}^1 \int_0^1 (x+y+z)\,dx\,dy\,dz$

Evaluate: $\int_0^2 \int_0^{\sqrt{y}} \int_0^{xy} 2xy^2 z \, dz \, dx \, dy$

The rectangular prism having as its two bases the triangle in the yz-plane cut out by the two axes and the line $y + z = 1$, and the corresponding triangle in the plane $x = 1$ obtained by adding 1 to the x-coordinate of each point in the first triangle. Supply limits for three different orders of integration:

(i) $\iiint dz\,dy\,dx$ (ii) $\iiint dx\,dz\,dy$ (iii) $\iiint dy\,dx\,dz$

Find the center of mass of the tetrahedron D in the first octant formed by the coordinate planes and the plane $x + y + z = 1$. Assume the density $\delta = 1$.

Find the volume underneath the surface $z = 16 - x^2 - 4y^2$ and over the xy-plane;